Vanchor Helm Board — sheet index\n1 power: battery in, protection, 5V/12V rails\n2 pi: Pi carrier, UARTs, I2C, display/fan power\n3 mcu: Pico 2 controller\n4 thrust: external 1500W controller hijack +

12–48V in; 1500W path stays off–board (external controller + contactor)
Pi 4/5 carrier + Pico 2: trolling motor thrust (digipot hijack) + AS5600 servo
vanchor–ng

Sheet: /
File: vanchor–helm.kicad_sch

Title: Vanchor Helm Board

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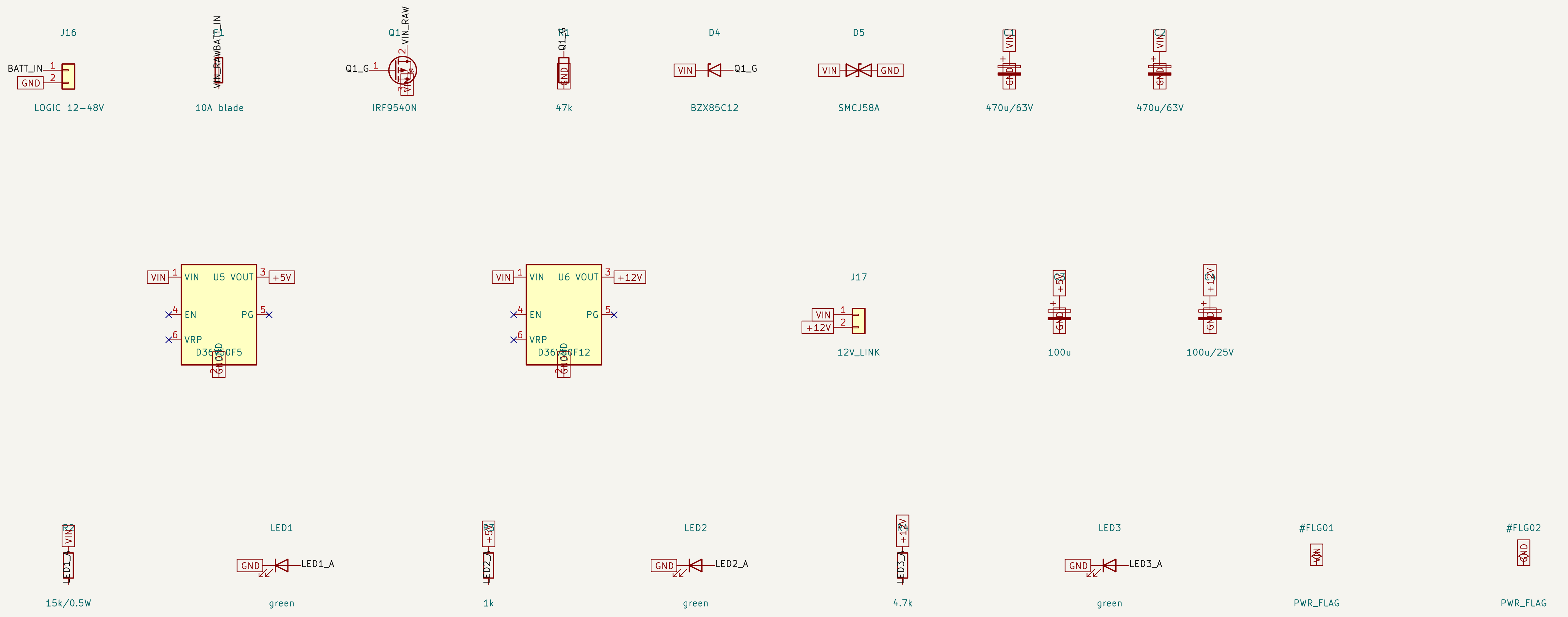
POWER: 12-48V battery in, protection, 5V/5A + 12V/4.5A rails

12V-battery boats: omit U6 (DNP) and bridge J17 with a wire link



Sheet: /power/
File: power.kicad_sch

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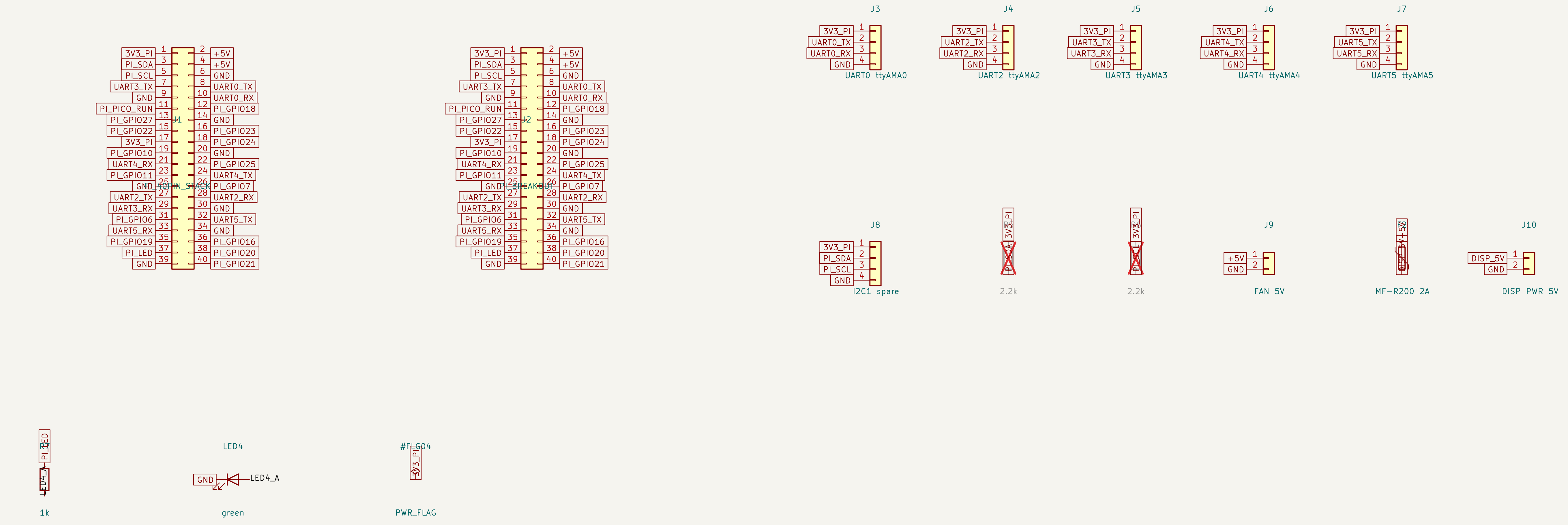
RASPBERRY PI 4/5 CARRIER: stacking header, UART breakouts, display power

DSI touchscreen: ribbon to Pi directly; 5V from J10. USB/HDMI on Pi itself.



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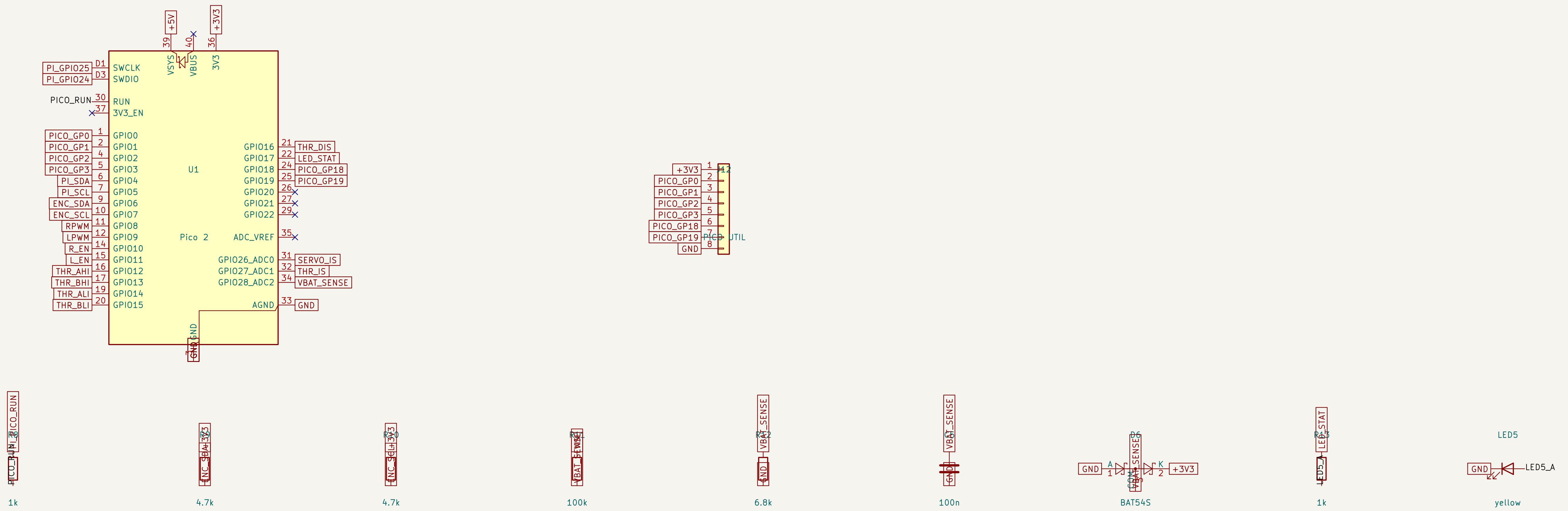
MCU: Pico 2 (RP2350) — I2C0 slave @0x42 to Pi, I2C1 master to AS5600

Firmware map: 0x00 CMD {pwm,dir,steer} / 0x10 STATUS {angle,ok,wrap,state,vbat,isense}; 800ms watchdog

Pico soldered directly (low profile, sits under the Pi). Pi reflashes it via SWD on GPIO24/25 (openocd).



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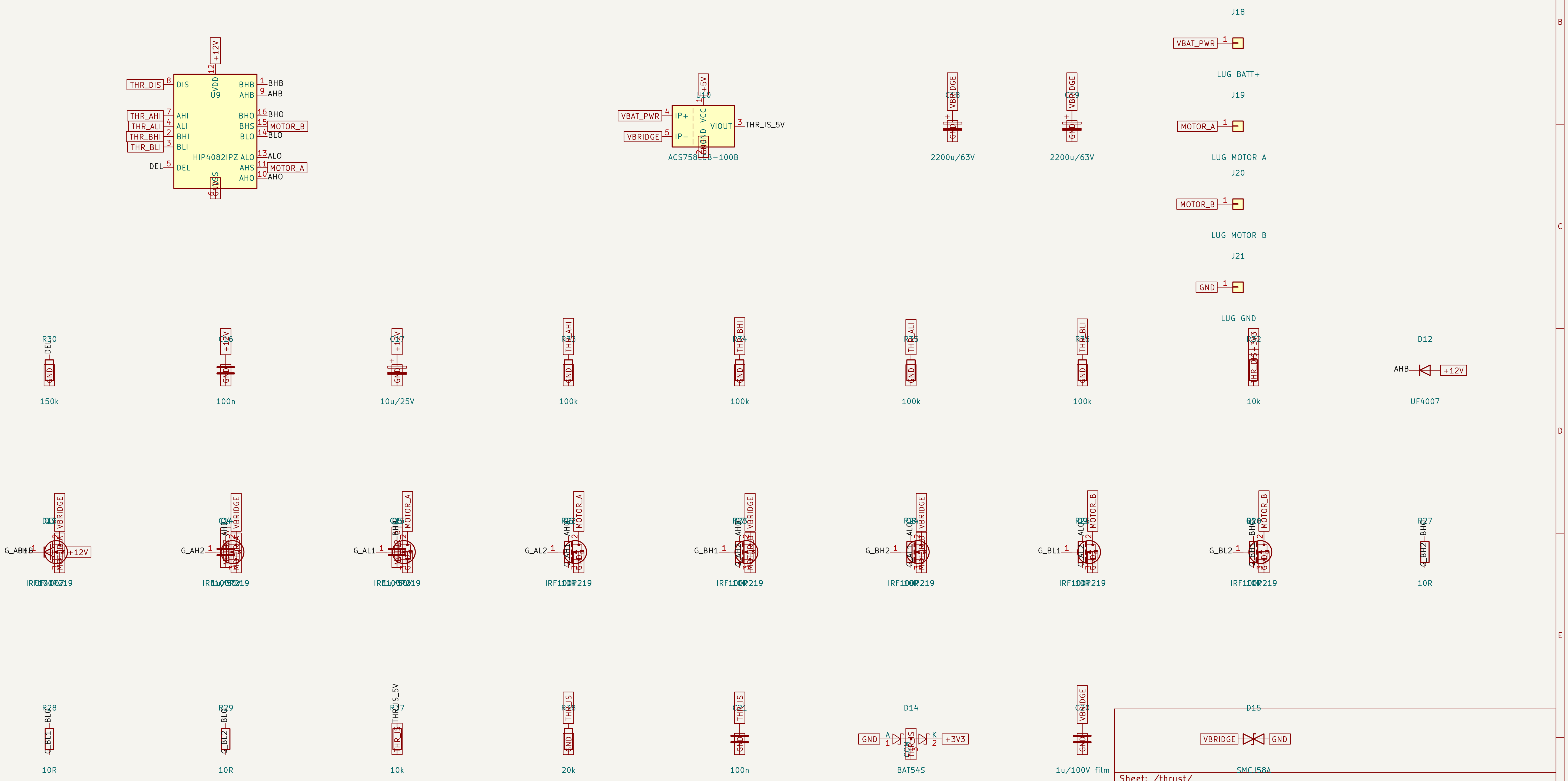
THRUST: on-board H-bridge, >=800W @ 12V (67A cont), fwd/rev, 12–48V bus

HIP4082 + 8x IRF100P219 (2 per switch) + ACS758-100B battery-leg sense; M5 lugs

Bridge disabled unless Pico drives it: 100k pull-downs on inputs, DIS pulled up to 3V3

Heatsink bar across the FET row required; airflow recommended above 40A continuous



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File: thrust.kicad_sch

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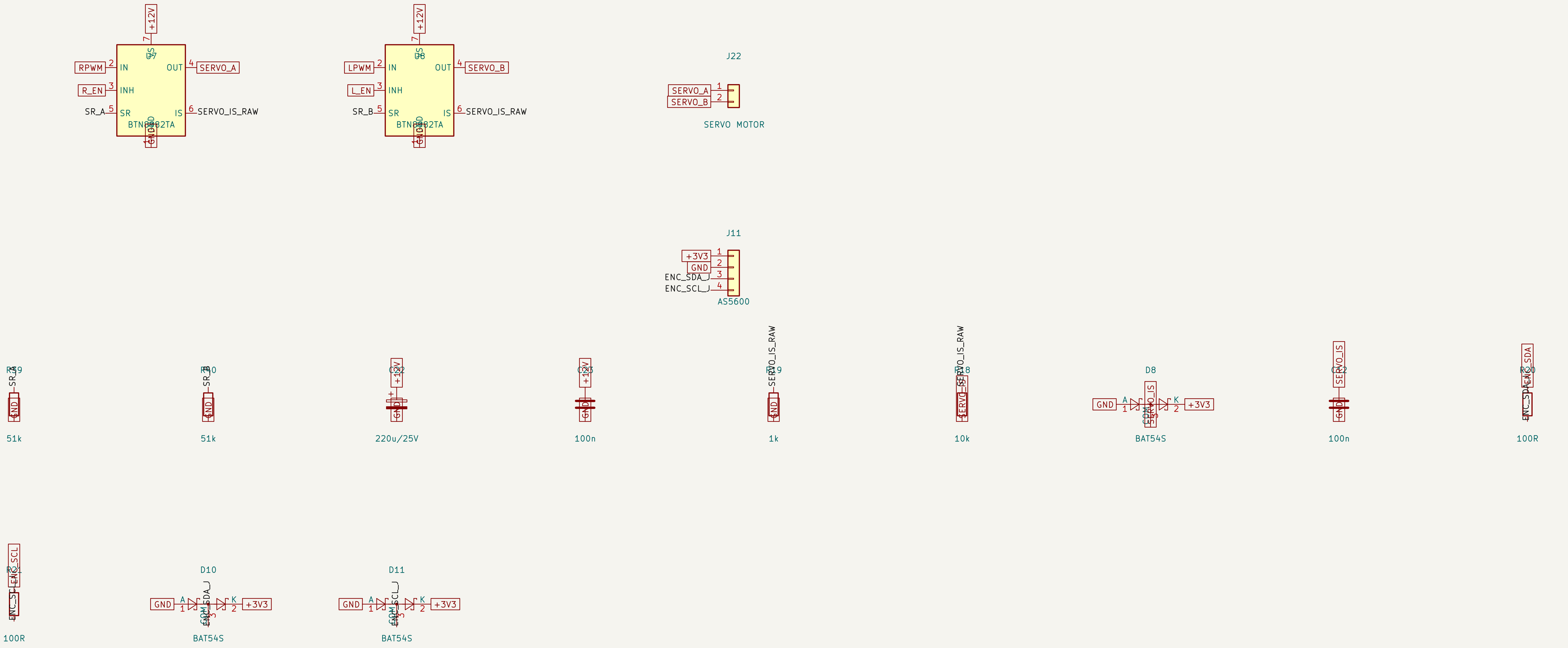
SERVO: on-board 2x BTN8982TA half-bridge (BTS7960-compatible interface), VM = +12V rail

AS5600 (in servo housing per vanchor-ng cad/) cabled to J11; keep lead <1m, 100kHz

24-48V boats: servo stall limited by BUCK2 (4.5A). 12V boats: J17 links VIN->12V, full stall current



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File: servo.kicad_sch

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KiCad E.D.A. 10.0.3 | Id: 6/6